

# AUTO TELLER MACHINE

System Analysis and Design

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# ATM System

## 1. Introduction

The **Automated Teller Machine (ATM) System** is a software-based banking interface that allows customers to perform financial transactions conveniently and securely. This project models the core functionalities of an ATM, including cash withdrawal, deposit, fund transfer, balance inquiry, receipt printing, and session termination. The system interacts with two external actors : the **User** (customer) and the **Bank System** (central banking server). The design follows UML standards and covers requirement analysis, process modeling, use case modeling, dynamic modeling (activity & sequence diagrams), and class diagram. The goal is to produce a complete, consistent, and well-documented system analysis and design deliverable.

## 2. Project Objectives

- **Who:** The system is designed for **bank customers** (users) who need self-service access to their accounts, and for **banking systems** that require secure transaction processing.
- **Why:** To provide a **secure, fast, and user-friendly** platform for basic banking operations without teller assistance, reducing queue times and enabling 24/7 access.
- **When used:** The system is used **whenever a customer needs to perform a cash withdrawal, deposit, transfer, or check balance** at an ATM, and optionally print a receipt or end the session.

### 3. Requirements Collection

All requirements for the ATM system were gathered using two complementary methods:

1. **Direct Observation:** We observed real ATM machines in public banks and retail locations. We noted the typical steps a user follows, the on-screen messages, error handling, and the sequence of interactions with the banking network.

**AI-Assisted User Simulation:** Since we have no real client or live user group, we used **generative AI** to simulate interviews with different types of users (e.g., frequent withdrawers, first-time users). The AI provided realistic scenarios, edge cases, and user expectations.

### 4. Functional Requirements

- The system shall allow the user to authenticate using a card and PIN.
- The system shall allow the user to select transaction if it was withdraw cash (must include authentication), or deposit cash (must include authentication), or transfer money between accounts (must include authentication), or check account balance (must include authentication).
- The system shall optionally print a receipt after withdrawal, check account balance, deposit, or transfer.
- The system shall allow the user to end the session .
- The system shall interact with the Bank System for all operations that modify or read the balance (withdraw, deposit, transfer, check balance).

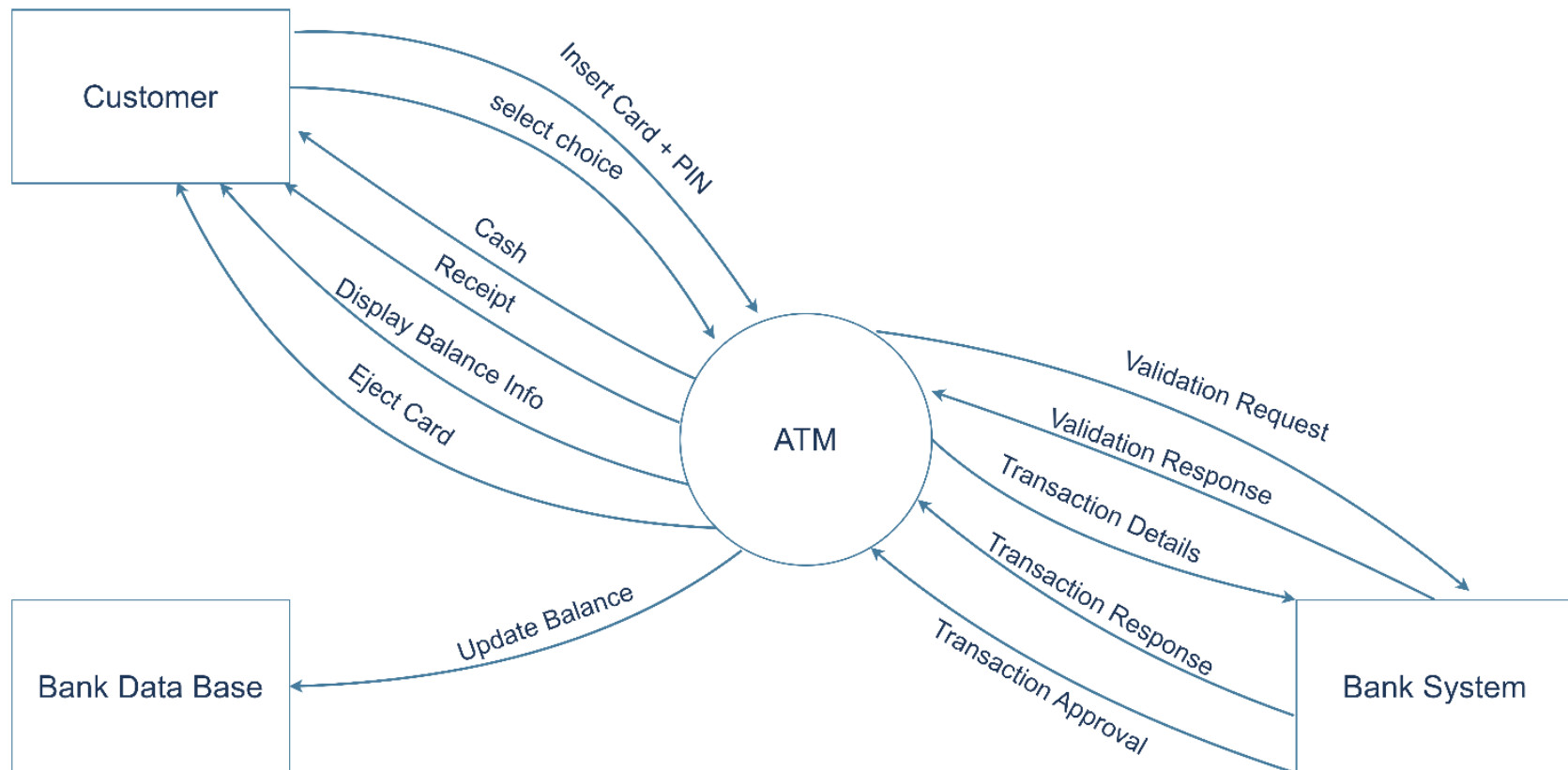
### 5. Non-functional Requirements

Authentication must be completed within 5 seconds (Performance), the system shall lock the session after 3 consecutive failed PIN attempts (Security), all communication with the Bank System must be encrypted (Security), the ATM screen interface must be intuitive – a first-time user should complete a withdrawal in less than 90 seconds (Usability), the system

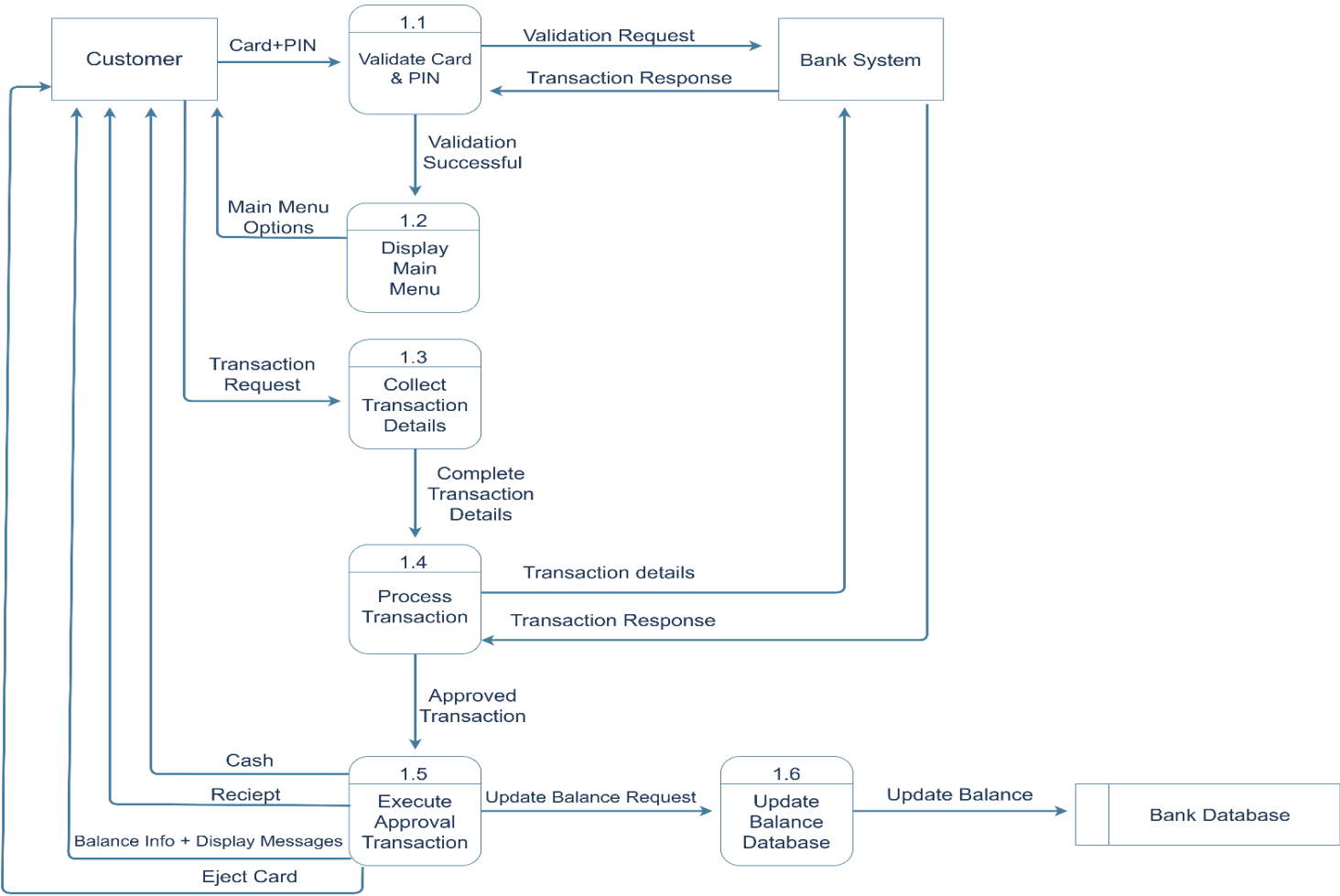
shall have 99.9% availability during planned banking hours (Reliability), receipt printing shall not require re-contacting the Bank System; the receipt only reflects already-confirmed transactions (Design constraint).

## 6.Process Models

### 6.1. Context Diagram



6.2.Data Flow Diagram



6.3. Use Case Diagram



## 6.4. Analysis of Use Cases

**\* Table 1- withdraw cash**

| Element               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case name</b>  | <b>Withdraw cash</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Actor</b>          | User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Preconditions</b>  | <ol style="list-style-type: none"><li>1. The customer has a valid bank card.</li><li>2. The ATM is online and has sufficient cash.</li><li>3. The bank system is available.</li></ol>                                                                                                                                                                                                                                                                                                                                         |
| <b>Postconditions</b> | <ol style="list-style-type: none"><li>1. The requested amount is deducted from the customer's balance.</li><li>2. Cash is dispensed to the customer.</li><li>3. A receipt is printed (optional).</li><li>4. The customer's balance is updated in the bank system.</li><li>5. The card is ejected.</li></ol>                                                                                                                                                                                                                   |
| <b>Main Flow</b>      | <ol style="list-style-type: none"><li>1. Customer inserts the card and enters PIN.</li><li>2. ATM validates card and PIN with the bank system.</li><li>3. ATM displays the main menu.</li><li>4. Customer selects "Withdraw Cash".</li><li>5. Customer enters the withdrawal amount.</li><li>6. ATM sends a transaction request (amount) to the bank system.</li><li>7. Bank system checks the customer's balance and approves the transaction.</li><li>8. ATM dispenses the cash.</li><li>9. ATM prints a receipt.</li></ol> |

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|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>10. ATM sends an update balance request to the bank system.</p> <p>11. Bank system updates the balance.</p> <p>12. ATM ejects the card.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Alternative Flows</b> | <p><b>Alt.1 – Invalid PIN</b></p> <ol style="list-style-type: none"> <li>1. At step 2 of Main Flow, the bank system returns "invalid PIN".</li> <li>2. ATM displays "Wrong PIN. Please try again."</li> <li>3. Increment retry counter.</li> <li>4. If retries &lt; 3, return to step 1 of Main Flow (re-enter PIN).</li> <li>5. If retries = 3, ATM displays "Card retained. Contact your bank." and ejects/retains the card. Use case ends.</li> </ol> <p><b>Alt.2 – Insufficient Balance</b></p> <ol style="list-style-type: none"> <li>1. At step 7 of Main Flow, the bank system returns "insufficient balance".</li> <li>2. ATM displays "Insufficient balance. Please enter a smaller amount."</li> <li>3. Return to step 5 of Main Flow (enter amount again) or return to main menu (design choice).</li> </ol> <p><b>Alt.3 – Cancel Transaction</b></p> <ol style="list-style-type: none"> <li>1. At any step before cash dispensed, customer presses "Cancel".</li> <li>2. ATM returns to main menu or ejects card (depending on policy).</li> </ol> <p><b>Alt.4 – ATM out of cash</b></p> <ol style="list-style-type: none"> <li>1. Before step 8, ATM detects no cash.</li> <li>2. ATM displays "Out of service" and ejects card. Use case ends.</li> </ol> |



**\* Table 2-Check Balance**

| Element        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use case name  | Check balance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Actor          | User                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Preconditions  | <ol style="list-style-type: none"><li>1. The customer has a valid bank card.</li><li>2. The ATM is online.</li><li>3. The bank system is available.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Postconditions | <ol style="list-style-type: none"><li>1. The current balance is displayed on the ATM screen.</li><li>2. Optionally, a receipt with the balance is printed.</li><li>3. The ATM returns to the main menu.</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Main Flow      | <ol style="list-style-type: none"><li>1. Customer inserts the card and enters PIN (or PIN already validated in a previous use case).</li><li>2. ATM validates card and PIN with the bank system.</li><li>3. ATM displays the main menu.</li><li>4. Customer selects "Check Balance".</li><li>5. ATM sends a balance inquiry request to the bank system.</li><li>6. Bank system retrieves the current balance.</li><li>7. Bank system returns the balance to the ATM.</li><li>8. ATM displays the balance on the screen.</li><li>9. ATM asks if customer wants a printed receipt.</li><li>10. Customer chooses yes/no.</li><li>11. If yes, ATM prints the balance receipt.</li><li>12. ATM returns to the main menu.</li></ol> |

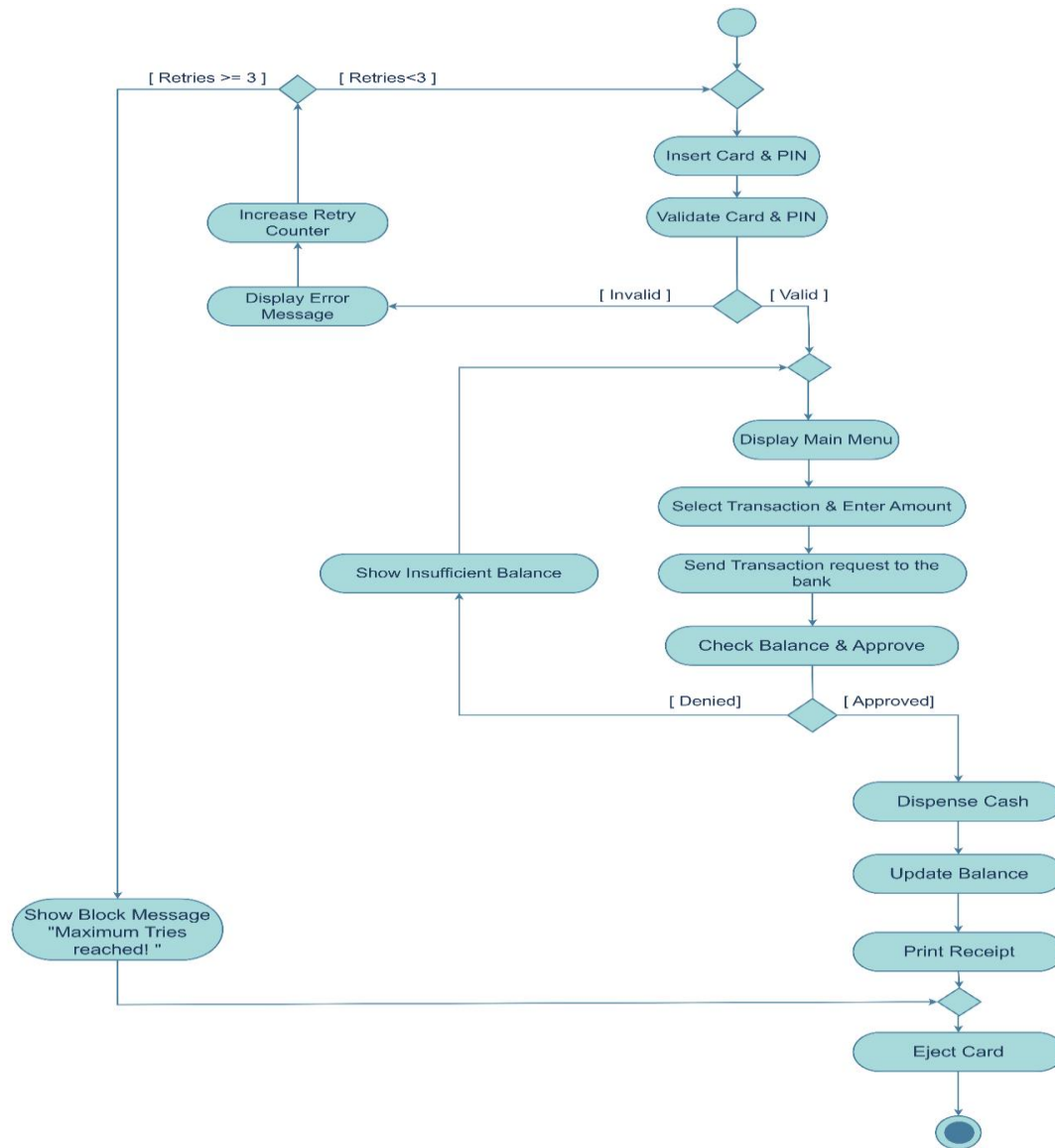
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| <b>Alternative Flows</b> | <p><b>Alt.1 – Invalid PIN</b><br/>Same as in Withdraw Cash (A1).</p> <p><b>Alt.2 – Bank system unavailable</b><br/>1. At step 6, the bank system does not respond or returns an error.<br/>2. ATM displays "Service unavailable. Please try again later."<br/>3. ATM ejects the card. Use case ends.</p> <p><b>Alt.3 – Cancel</b><br/>Customer presses "Cancel" before balance is displayed → ATM returns to main menu.</p> |

\* Table 3: Table 3: Validate Card & PIN

| Element               | Description                                                                                                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use case name</b>  | <b>Table 3: Validate Card &amp; PIN</b>                                                                                                                                                                                               |
| <b>Actor</b>          | User                                                                                                                                                                                                                                  |
| <b>Preconditions</b>  | 1. Customer has inserted a bank card into the ATM.<br>2. The ATM is powered on and online.                                                                                                                                            |
| <b>Postconditions</b> | <b>Success:</b> The ATM displays the main menu (a valid session is established).<br><b>Failure:</b> After 3 incorrect attempts, the card is retained/ejected and session ends.                                                        |
| <b>Main Flow</b>      | 1. Customer inserts the card into the ATM.<br>2. ATM prompts "Enter your PIN".<br>3. Customer enters the 4-digit PIN.<br>4. ATM sends the card number and encrypted PIN to the bank system.<br>5. Bank system checks the credentials. |

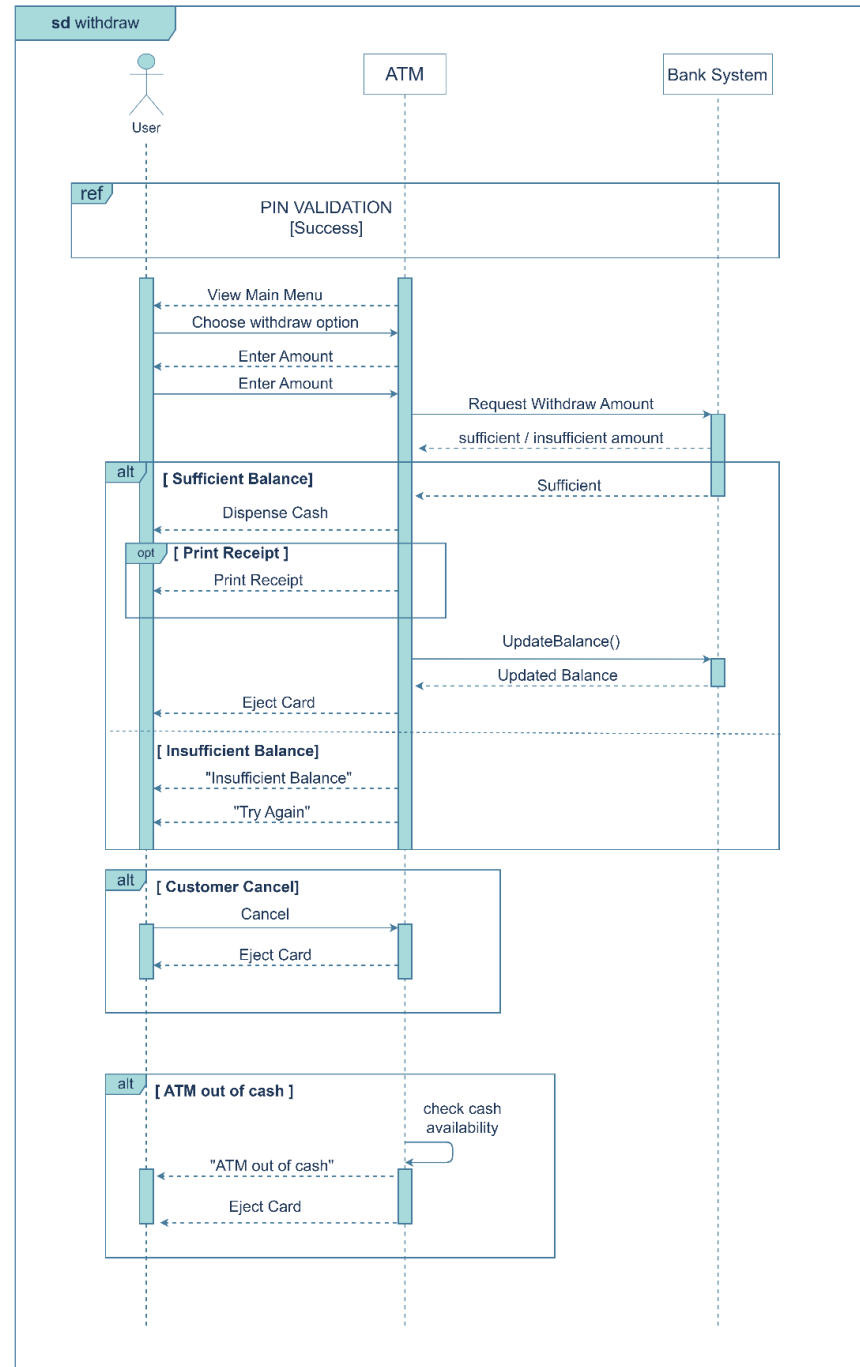
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|                          | 6. Bank system returns "valid".<br>7. ATM displays the main menu.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Alternative Flows</b> | <p><b>Alt.1 – Invalid PIN (with retries)</b></p> <ol style="list-style-type: none"> <li>1. At step 6, bank system returns "invalid".</li> <li>2. ATM displays "Wrong PIN. You have X attempts left."</li> <li>3. Increment retry counter.</li> <li>4. If retries &lt; 3, go back to step 2 of Main Flow (re-enter PIN).</li> <li>5. If retries = 3, ATM displays "Too many invalid attempts. Card retained." Then ATM retains the card (or ejects it depending on bank policy). Use case ends.</li> </ol> <p><b>Alt.2 – Expired or blocked card</b></p> <ol style="list-style-type: none"> <li>1. Bank system returns "card expired" or "card blocked".</li> <li>2. ATM displays "Card not valid. Contact your bank."</li> <li>3. ATM ejects the card. Use case ends.</li> </ol> <p><b>Alt.3 – Communication failure</b></p> <ol style="list-style-type: none"> <li>1. ATM cannot connect to bank system.</li> <li>2. ATM displays "Network error. Please try again later."</li> <li>3. ATM ejects the card. Use case ends.</li> </ol> |

## 6.5.Activity Diagram

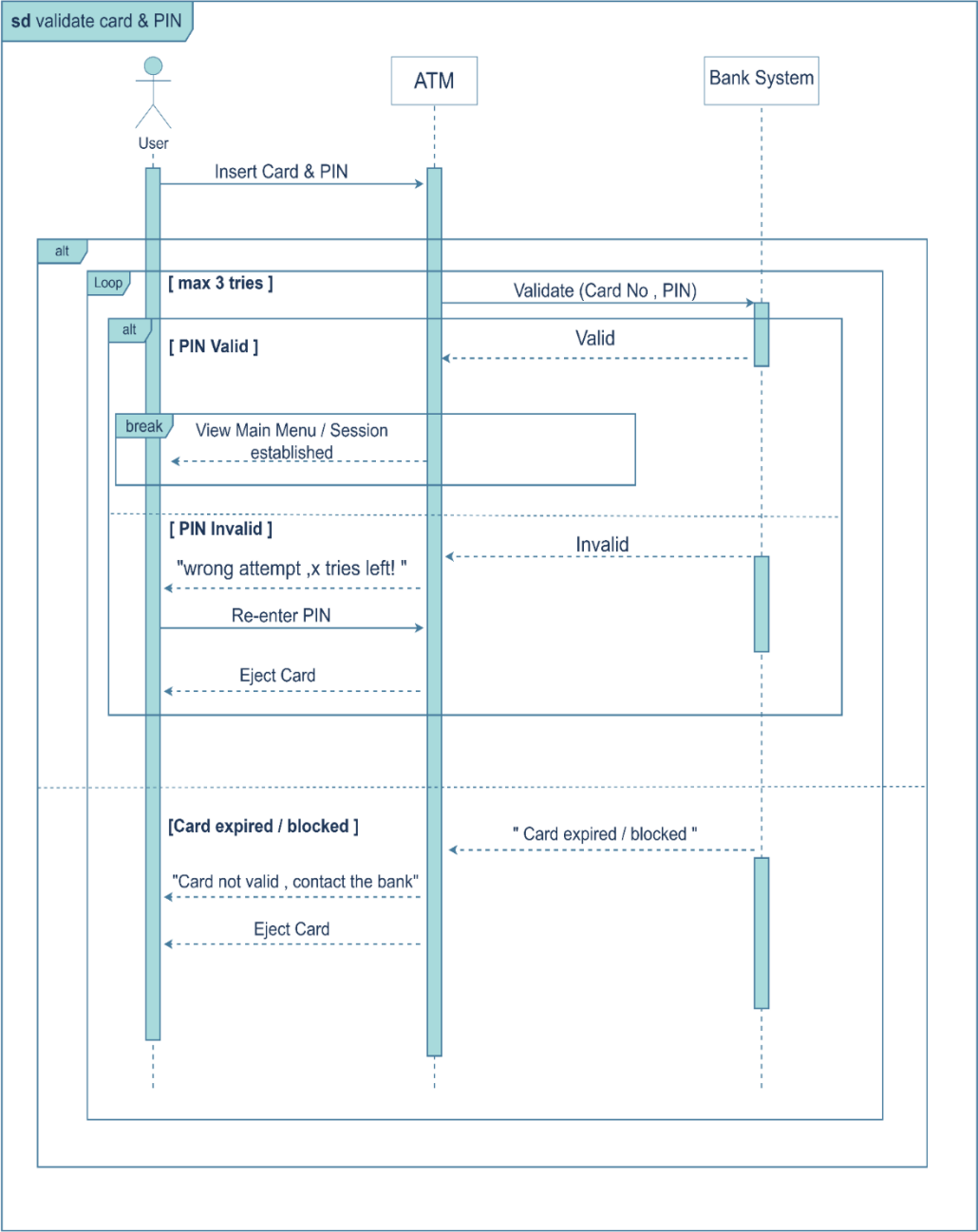


## 6.6. Sequence Diagrams

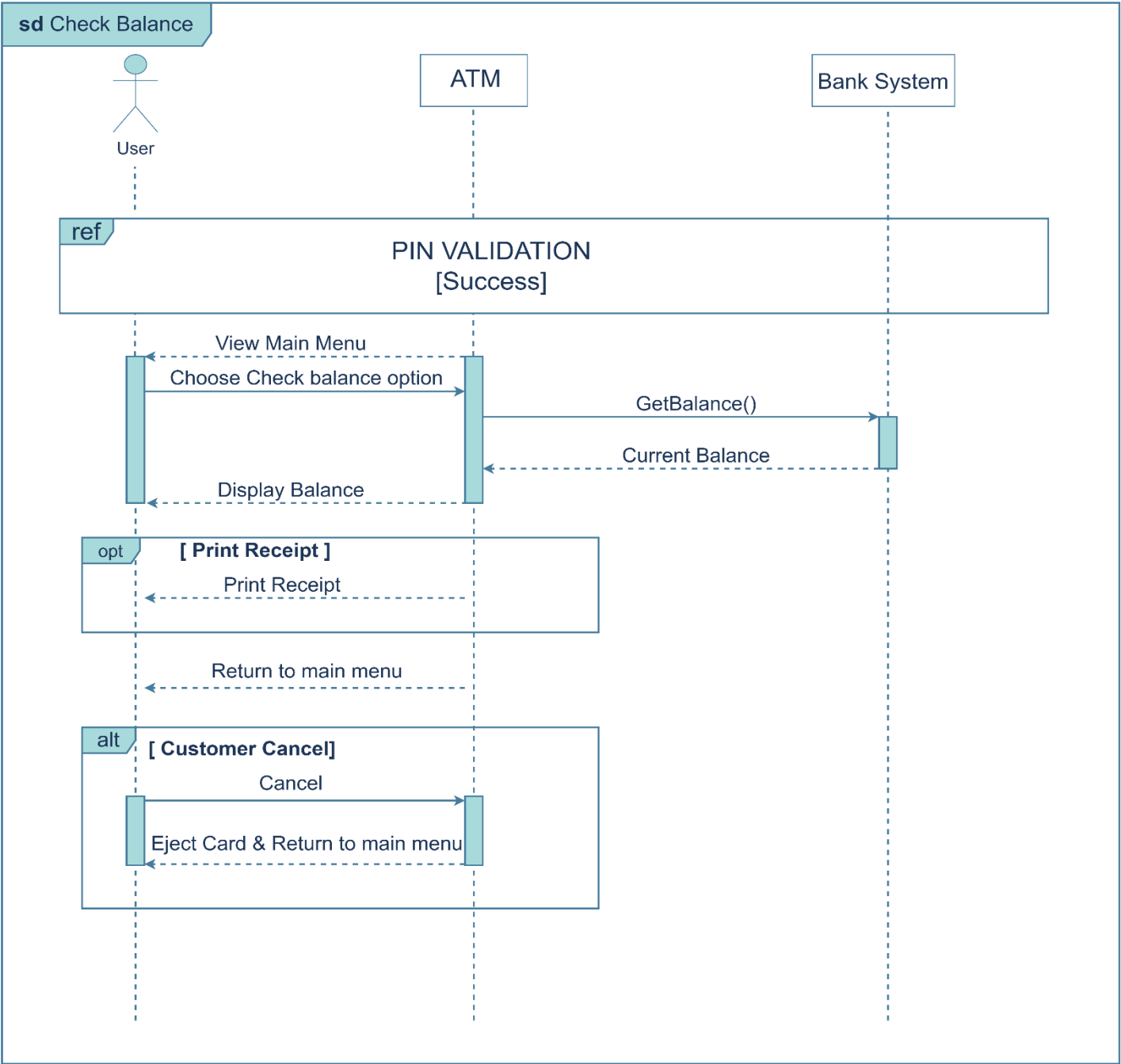
### 6.6.1. Withdrawal Sequence Diagram



6.6.2. Validation Card & PIN Sequence Diagram



6.6.3. Check Balance Sequence Diagram



# 6.7. Class Diagram

